

Mathematical Thinking

Module 1: Introduction to Mathematical Thinking

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2026 年 7 月 28 日

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目录

Knowledge Map

Topic	Core Question
Mathematical Thinking	What is mathematics really about? Why is it more than computation?
Precise Language	Why does math require more precision than everyday language?
Logical Connectives	How do AND, OR, NOT, IF...THEN combine propositions?
Quantifiers	How do we express 'for all' and 'there exists'? What are the negation rules?
Set Theory Basics	How are sets, empty set, subsets, unions, and intersections defined?

0. Overview

This unit introduces the mindset of mathematical thinking. Unlike school math, which focuses on procedures and computation, this course emphasizes **understanding the nature of mathematics**—a precise, rigorous, and abstract way of thinking developed over 3,000 years.

Core Objectives

- Transition from "school math" to "university math" thinking
- Understand that mathematics is the study of **patterns**
- Master the importance of precise language in mathematical reasoning
- Learn to use logical connectives and quantifiers for rigorous argumentation

1. 第-0-讲-欢迎辞

Review: 第-0-讲-欢迎辞 (8:15)

Keith Devlin opens with an analogy: if mathematics were the automotive world, **school math is learning to drive**, while this course teaches you **how the engine works**.

- Mathematical thinking is a mental ability developed over 3,000 years
- This course teaches no new formulas or computational techniques
- The goal is to help you transition from high school to university-level math
- Key distinction: school math asks **how**; university math asks **why**

Practical advice: Don't expect to solve every assignment problem in one sitting. What matters is **attempting each question**, not meeting a deadline.

集合论基础速查

01_ 第-0-讲-欢迎辞 _Supplement__Set_Theory_.pdf

内容摘要

- 集合论是后续课程的数学语言基础。本补充材料在课程后期才会正式用到，但建议先通读一遍建立整体印象。
- **集合**是由确定对象组成的整体。表示法：列举法 $\{1, 2, 3\}$ 或描述法 $\{x \in \mathbb{N} \mid x < 4\}$ 。**空集** $\emptyset$ 是不含任何元素的集合，注意 $\emptyset \neq \{\emptyset\}$ 。
- 核心关系与运算：**属于** $x \in A$ ；**子集** $A \subseteq B$ (A 的所有元素都在 B 中)；**并集** $A \cup B$ ；**交集** $A \cap B$ ；**补集** A^c 。这些运算满足交换律、结合律和分配律。

背景阅读：什么是数学？

01_ 第-0-讲-欢迎辞 _Background_Reading_.pdf

内容摘要

- 数学不仅仅是学习如何解方程、如何计算。学校把大量时间花在教授各种解题步骤上，却很少告诉学生这门学科到底在研究什么。这就像用“把球踢进球门的一系列动作”来解释足球——虽然没说错，但完全丢失了大局。
- 数学的核心是研究各种**模式** (patterns)：计数模式、测量模式、对称模式、运动模式、变化模式、空间模式等。数学家的真正工作是发现、描述、解释和证明这些模式。
- 数学符号（如代数表达式、公式、几何图形）不是故弄玄虚，而是精确描述抽象模式的必要工具——正如地图之于地形、乐谱之于音乐。掌握符号语言是进入数学思维的第一步。
- 数学的历史可追溯到古希腊（约公元前 6 世纪），泰勒斯和毕达哥拉斯首次提出“精确陈述的命题可以用形式论证来证明”，这标志着**定理**的诞生，也是数学区别于其他学科的根本特征。

2. 第-1-讲-介绍材料

Review: 第-1-讲-介绍材料 (27:59)

This lecture answers a fundamental question: **What is mathematics, really?**

Devlin points out that schools spend years teaching procedures but rarely explain what the subject is about. It's like explaining soccer by saying "it's a series of maneuvers to get the ball into the goal"—not wrong, but missing the big picture.

The Nature of Mathematics Mathematics is the study of **patterns**:

- Counting patterns (arithmetic)
- Measurement patterns (geometry)
- Symmetry patterns (group theory)
- Motion patterns (calculus)
- Change patterns (analysis)
- Randomness patterns (probability)

Mathematical notation is not pretension—it is a **necessary tool** for describing abstract patterns precisely, just as maps are for terrain and musical scores are for sound.

Historical milestone: Around the 6th century BCE, Thales and Pythagoras first proposed that precisely stated assertions could be proved by formal arguments—marking the birth of the **theorem**.

3. 第-2-讲-逻辑组合器

Review: 第-2-讲-逻辑组合器 (26:08)

This lecture introduces the core tools of mathematical reasoning: **logical connectives**.

Everyday language uses "and," "or," and "if...then" loosely. Mathematics requires **precise, unambiguous** definitions:

Five Basic Logical Connectives

1. **Conjunction (AND):** $P \wedge Q$ — true iff both P and Q are true
2. **Disjunction (OR):** $P \vee Q$ — true if at least one is true (inclusive OR!)
3. **Negation (NOT):** $\neg P$ — flips the truth value
4. **Conditional (IF...THEN):** $P \Rightarrow Q$ — false only when P is true and Q is false
5. **Biconditional (IFF):** $P \Leftrightarrow Q$ — true when P and Q have the same truth value

Common pitfall: In everyday English, "or" often means "one or the other but not both" (exclusive OR). In mathematics, $\vee$ includes the case where both are true.

4. 作业-1-的教程

Review: 作业-1-的教程 (2:30)

This tutorial covers Assignment 1, focusing on **Euclid's proof that there are infinitely many primes**.

Core argument:

1. Assume primes are finite: $p_1, p_2, \dots, p_n$

2. Construct $N = p_1 \times p_2 \times \cdots \times p_n + 1$
3. N leaves remainder 1 when divided by any p_i , so no known prime divides it
4. Therefore N is either a new prime itself, or has a prime factor not in the list
5. Either way, this contradicts "primes are finite"

Assignment tip: Question 8 asks "how to prove $p_1 p_2 \cdots p_n + 1$ is not always prime"—find a counterexample (e.g., $2 \times 3 \times 5 \times 7 \times 11 \times 13 + 1 = 30031 = 59 \times 509$).

5. 作业-2-的教程

Review: 作业-2-的教程 (9:21)

This tutorial covers Assignment 2: **simplifying logical expressions**.

Core technique: identify **superfluous conjuncts**:

- If $x < 4$, then $x < 6$ is automatically true — the second condition is redundant
- If $y^2 < 9$ (i.e., $-3 < y < 3$), then $y < 4$ is automatic — the first condition is redundant
- $7 \leq p < 12$ can be written concisely as a chained inequality

Method: Draw each condition as an interval on the number line, take the intersection, then check which condition is fully contained by another.

6. Key Takeaways

1. Mathematics is the study of patterns, not just computation.
2. Precise language is the foundation of mathematical reasoning—ambiguity is unacceptable.
3. Logical connectives (AND/OR/NOT/IF...THEN) have strict definitions; don't rely on everyday intuition.
4. Quantifier negation: $\forall$ becomes $\exists$, $\exists$ becomes $\forall$, and the inner proposition is negated.
5. The heart of proof is logical argument, not example verification.

7. Exercises

A. Multiple Choice (10 questions)

1. What does mathematics primarily study?
A. Number crunching B. Patterns C. Physical measurement D. Computer programming
2. $P \vee Q$ is false when:
A. P false, Q true B. P true, Q false C. Both P and Q are false D. Both P and Q are true
3. $P \Rightarrow Q$ is false only when:
A. P false, Q false B. P false, Q true C. P true, Q false D. P true, Q true

4. $\neg\forall x P(x)$ is equivalent to:
A. $\forall x \neg P(x)$ B. $\exists x \neg P(x)$ C. $\exists x P(x)$ D. $\neg\exists x P(x)$
5. Euclid's proof of infinitely many primes uses:
A. Direct proof B. Mathematical induction C. Proof by contradiction D. Constructive proof
6. Which correctly represents the empty set?
A. $\{0\}$ B. $\{\}$ C. $\{\emptyset\}$ D. 0
7. If $x < 4$, then $x < 6$ is:
A. An independent condition B. A superfluous condition C. A contradictory condition D. A necessary and sufficient condition
8. The main difference between mathematical OR and everyday 'or' is:
A. Math OR excludes both being true B. Math OR includes both being true C. No difference D. Math OR only applies to numbers
9. $A \subseteq B$ means:
A. A equals B B. Every element of A is in B C. Every element of B is in A D. A and B share no elements
10. The birth of the theorem marked:
A. Math started using numbers B. Precise assertions can be proved by formal arguments C. Algebra separated from geometry D. The invention of calculus

B. True/False (5 questions)

1. In mathematics, $P \vee Q$ is still true when both P and Q are true. ()
2. $P \Rightarrow Q$ is always false when P is false. ()
3. $\neg\exists x P(x)$ is equivalent to $\forall x \neg P(x)$. ()
4. The empty set $\emptyset$ and $\{\emptyset\}$ are the same set. ()
5. A single example can prove a universal statement. ()

C. Short Answer (2 questions)

1. Explain why mathematics requires more precise language than everyday speech. Give an example where everyday ambiguity leads to misunderstanding.
2. Outline Euclid's proof of infinitely many primes using proof by contradiction. Clearly state the assumption, construction, and contradiction.

8. Answer Key

Multiple Choice

1.B 2.C 3.C 4.B 5.C 6.B 7.B 8.B 9.B 10.B

True/False

1. True 2. False 3. True 4. False 5. False

Short Answer Key Points

1. Everyday 'or' often means exclusive (one or the other), but $\vee$ includes both; everyday 'if' implies causation, but $\Rightarrow$ only checks truth values. Precision eliminates ambiguity.
2. Assume primes are finite $p_1, \dots, p_n$; construct $N = p_1 p_2 \cdots p_n + 1$; no p_i divides N , so either N is prime or has a new prime factor—contradiction.